

Deep Imbalanced Separation Network: A Holistic Fault Detection Framework Considering Class-Imbalance and Partial Label-Unknown

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Abstract—The challenges of class-imbalance and partially unknown training labels often arise in fault detection tasks. When these two problems occur simultaneously, existing imbalanced classification methods cannot be directly used due to the absence of the label, and the class-imbalance would lead to severe bias prediction. In this study, we proposed a novel deep imbalance separation network (deepImSN) framework that is capable of dealing with fault detection problems with both class imbalance and partially unknown labels. This framework integrates the one-class learning concept into the positive-unlabeled (PU) learning theory for the first time. It alleviates the bias of the class-imbalance while making full use of the limited label information in the PU set to optimize the feature space and guide model training. The proposed deepImSN is designed to be used in different scenarios. It can accurately complete the fault detection task whether only part of fault samples or normal samples are labeled, and the class-prior is known or unknown. Experimental results on real-world problems, such as high-speed rail wheels fault inspection and wafer map fault detection, demonstrate that deepImSN outperforms existing methods in various experimental conditions.

Index Terms—Class-imbalance, fault detection, label unknown, positive-unlabeled (PU) learning.

I. INTRODUCTION

FAULT detection is an indispensable part of intelligent management in industrial systems [1]. It can detect potential problems in the system early on, prevent equipment from serious malfunctions causing production interruptions, and improve the safety and stability of system operation [2], [3]. In practical fault detection problems, there are usually two challenges. The first is

the issue of class-imbalance due to the scarcity of fault samples. Taking high-speed trains as an example, China's high-speed trains have achieved an average failure rate of less than 0.43 per million kilometers [4], which makes it difficult to collect enough failure data. The class-imbalance can lead to serious training bias [5], [6]. The model will tend to classify samples into the majority class, making it difficult to capture a small amount of fault information in the training set [7]. The second challenge is the high labeling cost of samples, making it difficult to obtain a large number of accurately labeled samples for training [8]. For example, for blade cracking faults in wind turbines, the data acquisition system can obtain a large amount of sensor data, but it is difficult for staff to find early faults and accurately label them during the operation of the wind turbine [9].

Regarding the issue of class-imbalance, existing research can be classified into the following three categories [10].

- 1) *Resampling methods*: By oversampling the minority class samples [11] or undersampling the majority class samples [12] to balance the distribution of the dataset before training, traditional methods can be used to train detection models.
- 2) *Model design-based methods*: By calibrating the model learning objective and designing models based on the assumption of imbalanced data, models can be directly trained. A typical example is cost-sensitive learning [5], [13], which penalizes misclassification of minority class samples more severely to mitigate model bias caused by imbalanced distribution.
- 3) *Ensemble learning*: Ensemble learning combines multiple base classifiers to obtain more reliable classification results by combining resampling methods or model design-based methods [14], [15].

According to the label information contained in the training set, fault detection methods can be divided into unsupervised learning, semisupervised learning, and supervised learning. Unsupervised learning does not use any label information, so the detection accuracy is usually poor [16]. Supervised learning requires that all samples in the training set must be accurately labeled, which is often difficult to meet the requirements in practical applications. Compared to these two, semisupervised learning methods can better balance label costs and classification accuracy [17]. Furthermore, since labeling fault samples in industrial systems is often more difficult than labeling normal

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samples, fault detection models based on positive-unlabeled (PU) learning have started to develop rapidly [18], [19]. PU detection models only require labeling part of normal samples in the training set, making them more suitable for situations where labeling samples is difficult. For example, Qian et al. [9] proposed a PU learning-based wind turbine fault detection method, which achieved classification performance close to that of supervised learning using only a small amount of labeled sample.

In practical industrial applications, the two aforementioned problems often occur simultaneously. When both class imbalance and partial label unknown exist, existing methods for class-imbalance cannot be directly applied to alleviate the class imbalance due to unknown labels. Existing PU learning methods will face serious bias, and the model will have difficulty converging to an ideal result. Only a few studies have focused on how to solve classification problems when these two problems occur simultaneously. Chen et al. [20] proposed a cost-sensitive PU (CSPU) learning method. It assigned distinct weights to the losses caused by false negative and false positive samples, and employed double hinge loss to build CSPU algorithm under the framework of empirical risk minimization. Su et al. [21] proposed a general learning objective for PU learning targeting specially at imbalanced data. By this general learning objective, PU methods based on optimizing a consistent risk estimator can be adapted to conquer the imbalance.

These research works all assume that the labeled samples are the faulty samples (minority class). However, labeling normal samples is much simpler than labeling faulty samples in some fault detection problems [9]. If the labeled samples are normal samples (majority class), the problem will be largely different. It is necessary to establish a general framework to solve imbalanced PU problems in various class-imbalance situations. In addition, the abovementioned PU learning methods all require that the prior probability of positive class $\pi_p = P(y = +1)$ of the training set is known or can be accurately estimated [22], which is equivalent to the situation that the imbalance ratio (IR) of the training set is known [9]. In practice, given the unlabeled training samples this condition obviously cannot hold, and thus, the existing PU methods may not be directly applicable.

Based on the abovementioned analysis, we propose a deep imbalanced separation network (deepImSN) framework for fault detection problems under the coexistence of class-imbalance and partially unknown labels. Inspired by unsupervised learning methods, such as support vector domain description (SVDD) [23], the framework uses hyperspheres as classification boundaries in the feature space to encompass as many majority class samples as possible. At the same time, it leverages the small amount of label information in the PU training set to guide model training and optimize the feature space to improve separability, ultimately achieving a good classification performance. Moreover, deepImSN optimizes a deep neural network (DNN) to map samples into a feature space instead of traditional kernel methods in SVDD. Prior to formal training, the model increases information density through unsupervised pretraining, extracting feature spaces with high relevance for classification. The framework is very flexible and has designed corresponding model adjustment

and optimization strategies for different scenarios, including training sets with labeled majority or minority class samples, known or unknown class prior. Therefore, it is applicable to various imbalanced PU classification problem scenarios.

The main innovations of this study can be summarized as follows.

- 1) A novel deepImSN framework is proposed, which can solve the fault detection problem when the dataset is class-imbalanced and partial labels are unknown. The proposed method combines the ideas of one-class learning and PU learning. The training of deepImSN only needs a partial normal samples label or a partial faulty samples label. The unique optimization objective setting can effectively avoid the learning bias caused by class-imbalance.
- 2) Compared with existing unsupervised separation methods, the proposed method can make full use of the limited label information in the PU training set to optimize the feature space and guide model training. By appropriately adjusting the optimization objective, the proposed framework can be applied in situations when the class prior is unknown.
- 3) The framework was validated on two industrial fault detection tasks, high-speed train wheel fault detection and wafer map defect detection. The proposed method outperformed benchmark methods under different class IRs and different labeled proportions. Particularly, when only a part of normal samples was labeled, other benchmark methods showed a poor performance while the proposed method maintained a high level of accuracy.

The rest of this article is organized as follows. Section II presents the problem definition and unsupervised separation methods. Section III introduces the proposed deepImSN framework in detail. Section IV verifies the feasibility and effectiveness of the proposed framework on two real-world cases. Finally, Section V concludes this article.

II. PRELIMINARIES

A. Problem Definition

The fault detection problem can be defined as a binary classification problem. Since the training set in this study is a partially labeled PU set, we use a triplet (x, y, s) for representation. $x \in \mathbb{R}^d$, $y \in \{\pm 1\}$ and $s \in \{\pm 1\}$ are the input feature space, the true class and label, respectively. The true class y is a hidden variable, and the model can only use the label s for training. For sample x_i , $s_i = +1$ means that the sample is labeled as positive sample, $y_i = +1$. $s_i = -1$ indicates that we do not know the true class of the sample, $y_i \in \{\pm 1\}$. Hence, there are the following three types of samples in the PU set.

- 1) *Positive labeled sample*: $\{x_i | s_i = +1, y_i = +1\}$. The size is n_{pl} .
- 2) *PU sample*: $\{x_i | s_i = -1, y_i = +1\}$. The size is n_{pu} .
- 3) *Negative unlabeled sample*: $\{x_i | s_i = -1, y_i = -1\}$. The size is n_{nu} .

The distribution of the training set in this study is imbalanced. Depending on the class and number of labeled samples, the following three imbalanced scenarios may exist.

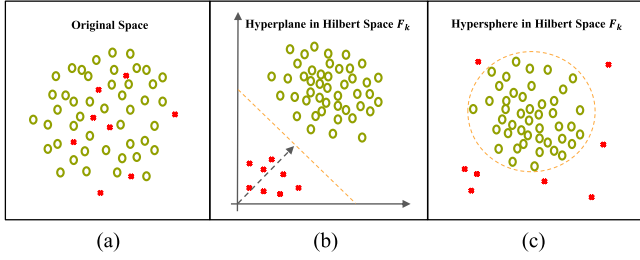


Fig. 1. Schematic diagrams of OC-SVM and SVDD [26]. (a) Dataset. (b) OC-SVM. (c) SVDD.

- a) $n_{pl} + n_{pu} \ll n_{nu}$. That is, the labeled samples are the minority class (fault samples).
- b) $n_{pl} + n_{pu} \gg n_{nu}$. That is, the labeled samples are the majority class (normal samples).
- c) $n_{pl} \ll n_{pu} + n_{nu}$. That is, the number of labeled samples is much smaller than the number of unlabeled samples.

The imbalance scenarios corresponding to (a), (c) or (b), (c) usually appear together, which brings huge deviations to the optimization of the model.

The objective of the training is to learn the potential mapping relationship from the feature x to the true class y based on the input feature x and label s .

B. Unsupervised Separation Methods

Unsupervised separation methods determine the boundary by examining the distribution of the training set without using the labels. These methods are less affected by class-imbalance and are able to handle imbalanced classification problems well in cases where class overlap is not obvious.

Typical unsupervised separation methods include one-class support vector machine (OC-SVM) and SVDD. Their schematic diagrams are shown in Fig. 1. Both methods are kernel-based. Let $\kappa : \mathcal{X} \times \mathcal{X} \rightarrow [0, +\infty)$ be a positive semi-definite kernel, F_κ is associated reproducing Kernel Hilbert space, $\phi_\kappa : \mathcal{X} \rightarrow F_\kappa$ is the associated feature mapping. For samples $x_i, x_j \in \mathcal{X}$, the kernel function is defined as [24]

$$\kappa(x_i, x_j) = \langle \phi_\kappa(x_i), \phi_\kappa(x_j) \rangle_{F_\kappa} \quad (1)$$

where $\langle \cdot, \cdot \rangle_{F_\kappa}$ is the dot product in Hilbert space F_κ .

1) One-Class Support Vector Machine: The objective of OC-SVM is to find a hyperplane ($w \cdot \phi_\kappa(x_i) - \rho = 0$) with maximum margin in the feature space that best separates the mapped data from the origin [25]. Assuming the training set is $D_n = \{x_1, x_2, \dots, x_n\}$, the objective of OC-SVM is [25]

$$\begin{aligned} \min_{w, \rho, \xi} \quad & \frac{1}{2} \|w\|^2 - \rho + \frac{1}{vn} \sum_{i=1}^n \xi_i \\ \text{s.t.} \quad & \langle w, \phi_\kappa(x_i) \rangle_{F_\kappa} \geq \rho - \xi_i, \xi_i \geq 0 \end{aligned} \quad (2)$$

where ρ is the distance from the hyperplane to the origin. ξ_i is slack variables of the soft margin. v is the weight parameter. The first and second terms of the cost function in (2) are to maximize the distance from the hyperplane to the origin, and the third term is the added slack variable to make a soft classification margin.

2) Support Vector Domain Description: SVDD is a method similar to the OC-SVM in which a hypersphere is used to separate the dataset instead of a hyperplane. The objective of SVDD is to find the smallest hypersphere with center $c \in F_\kappa$ and radius $R > 0$ that encloses the majority of the data in feature space F_κ . Assuming the training dataset is D_n , the objective of SVDD is given by [23]

$$\begin{aligned} \min_{R, c, \xi} \quad & R^2 + \frac{1}{vn} \sum_{i=1}^n \xi_i \\ \text{s.t.} \quad & \|\phi_\kappa(x_i) - c\|^2 \leq R^2 + \xi_i, \xi_i \geq 0 \end{aligned} \quad (3)$$

where ξ_i and v are the slack variables and weight parameter, respectively.

III. DEEPMNSN FRAMEWORK

A. Deep Separation Network Framework

Whether it is OC-SVM or SVDD, the optimization is based on the kernel method to obtain a feature space with better separability. However, choosing an appropriate kernel function is difficult and dataset-dependent in the application. DNNs are more suitable for data-driven methods compared with kernel methods. Their feature representation capabilities are strong enough. Various network structures can be flexibly adjusted according to data types. Therefore, we choose to adopt DNN instead of kernel function to build the framework [27]. For input feature space $\mathcal{X} \in \mathbb{R}^d$ and mapping feature space $\mathcal{Y} \in \mathbb{R}^{d'}$, let $\phi(\cdot, \mathbf{W}) : \mathcal{X} \rightarrow \mathcal{Y}$ be a neural network with $L \in \mathbb{Z}$ hidden layers and set of weights $\mathbf{W} = \{W^1, W^2, \dots, W^L\}$.

Unlike unsupervised learning, PU learning benefits from the labeling of a portion of positive class samples, which can guide the establishment of the correct boundary. Existing research on deep learning has shown that using task-specific label information to fine-tune the model can significantly improve model performance on specific tasks [28]. Therefore, in this research, how to use the label information of the partial positive class samples in the training set is the key to solving the problem. Especially when the dataset has a certain class overlap, it is necessary to adjust the loss function through the label information to guide the model to derive a high-dimensional feature space with better separability.

The optimization objective proposed in the framework can be divided into three parts shown as

$$\min_{\mathbf{W}} \hat{R}_{PU}(\mathbf{W}) = \hat{R}_U(\mathbf{W}) + \hat{R}_P(\mathbf{W}) + L_2(\mathbf{W}). \quad (4)$$

The first part is the misclassification empirical risk on the unlabeled set $\hat{R}_U(\mathbf{W})$, which is similar in design to unsupervised separation methods due to the lack of label information. It should be noted that this part needs to be adjusted according to the different class-imbalanced scenarios mentioned earlier. The second part is the misclassification of empirical risk on the positive set $\hat{R}_P(\mathbf{W})$, which mainly uses the supervised information to guide the model to construct an appropriate classification boundary. The final part is the regularization term of the model $L_2(\mathbf{W})$, which uses L_2 regularization in this study.

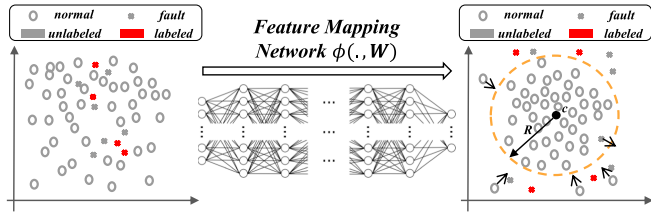


Fig. 2. Schematic diagrams of Scenario I (part of fault samples are labeled).

to reduce the risk of overfitting and improve the generalization performance of the model. Next, we will introduce in detail the design of the framework under the different class-imbalanced scenarios.

B. Scenario I: Part of Faulty Samples are Labeled

Assuming that the positive class is the minority class, which refers to the fault sample. In this case, only a portion of faulty samples are labeled, and the remaining samples are all unlabeled, that is, $n_{pl} + n_{pu} \ll n_{nu}$. Since the number of fault samples is much smaller than that of the normal samples, the number of labeled samples is also much smaller than that of the unlabeled samples, $n_l = n_{pl} \ll n_{pu} + n_{nu} = n_u$.

The training dataset is depicted in Fig. 2. Due to the effect of class imbalance, the vast unlabeled dataset are normal samples, while the labeled samples are all fault samples. Therefore, the first goal of the classifier is to separate the labeled samples from the unlabeled samples. In addition, since there are still a few fault samples in the unlabeled dataset, it is also necessary to separate them from the rest of the unlabeled dataset as much as possible. This task is completely consistent with the goal of the unsupervised separation method mentioned earlier.

To achieve the two goals mentioned above, the objective function of the model can be set as

$$\begin{aligned} \min_{\mathbf{W}, R, c} & R^2 + \frac{1}{n_u + 1} \sum_{i=1}^{n_u} \max \{0, \|\phi(x_i; \mathbf{W}) - c\|^2 - R^2\} \\ & - \frac{1}{n_l + 1} \sum_{i=n_u+1}^{n_u+n_l} \min \{0, \|\phi(x_i; \mathbf{W}) - c\|^2 - R^2\} \\ & + \frac{\lambda}{2} \sum_{l=1}^L \|\mathbf{W}^l\|^2. \end{aligned} \quad (5)$$

The essence of this objective function is using a hypersphere to include normal samples as much as possible while excluding fault samples. Minimizing R^2 in the first term is to minimize the volume of the hypersphere. The second term is the penalty for unlabeled samples lying outside the hypersphere, i.e., if samples' distance to the center $\|\phi(x_i; \mathbf{W}) - c\|^2$ is bigger than the radius R . This term plays a role similar to a soft-boundary SVDD [27], helping to separate fault samples within the unlabeled dataset. Since the labeled samples are all fault samples, we assume they are far away from the hyperplane. Hence, the third term is the penalty for labeled samples lying inside the hypersphere, i.e., if samples' distance to the center $\|\phi(x_i; \mathbf{W}) - c\|^2$ is smaller than

the radius R . This term, along with the second term, plays a role similar to supervised learning, separating the labeled samples from the unlabeled samples. The last term is the regularizer for the DNN parameters $\mathbf{W}$ with hyperparameter $\lambda > 0$, where $\|\cdot\|$ denotes the Euclidean norm.

Taking the average of each penalty term in the objective function is to avoid model bias caused by the class-imbalance of labeled and unlabeled samples. It should be noted that since neural network optimization is based on mini-batches, to avoid the denominator being zero when a batch contains only unlabeled or labeled samples, we added an additional one to the denominator in the objective function.

As both center c and radius R are meaningful and related to the entire dataset, treating them as free optimization variables easily lead to a trivial and uninformative solution. Therefore, it is necessary to design estimation methods for these two parameters based on the properties of the PU imbalanced dataset.

According to the case-control assumption, the entire unlabeled set is randomly sampled from the true distribution, so the initial center c of the hypersphere can be simply set as the mean of all the unlabeled data after feature mapping

$$c = \frac{1}{n_u} \sum_{i=1}^{n_u} \phi(x_i; \mathbf{W}). \quad (6)$$

If the class prior π_p is known, we can estimate the number of normal samples in the unlabeled dataset n_{nu} . After the feature mapping, normal samples are located near the center of the hypersphere while fault samples are far from it. Therefore, we can determine the radius R by the order statistic of the distance from the unlabeled samples to the center. We set R to satisfy the following equation:

$$\lceil (1 - \pi_p)(n_l + n_u) \rceil = \sum_{i=1}^{n_u} \mathbf{1}[\|\phi(x_i; \mathbf{W}) - c\| \leq R]. \quad (7)$$

If the class prior π_p is unknown, the left-hand side of (7) cannot be estimated. Due to the lack of prior information, it is difficult to accurately determine the radius of the hypersphere, which will lead to difficulties in model convergence of (5). In this case, we propose a simplified optimization objective

$$\begin{aligned} \min_{\mathbf{W}, c} & \frac{1}{n_u + 1} \sum_{i=1}^{n_u} \|\phi(x_i; \mathbf{W}) - c\|^2 \\ & - \frac{1}{n_l + 1} \sum_{i=1}^{n_l} \|\phi(x_i; \mathbf{W}) - c\|^2 \\ & + \lambda/2 \sum_{l=1}^L \|\mathbf{W}^l\|^2. \end{aligned} \quad (8)$$

Compared with (5), the simplified objective function penalizes the mean distance to the center of all unlabeled samples instead of only the unlabeled samples falling outside the hypersphere. It avoids introducing the radius parameter R . The essence of this objective function is to find a hypersphere that makes normal samples as close to the center as possible while fault samples are as far away from the center as possible. The first term is to minimize the average distance between the unlabeled samples

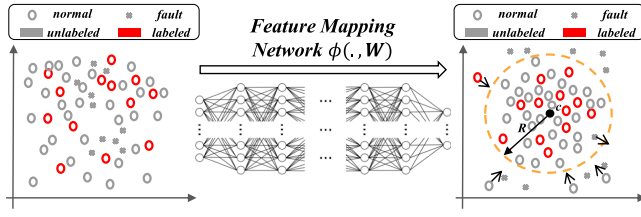


Fig. 3. Schematic diagrams of Scenarios II (part of normal samples are labeled).

and the center c . The second term is to maximize the average distance from the labeled samples to the center c . The regularizer term is the same as (5).

C. Scenario II: Part of Normal Samples are Labeled

Existing studies on PU imbalanced learning all assume the positive class is the minority class. However, in many real-world fault detection problems, labeling the fault samples (minority class) is much more difficult compared to labeling normal samples (majority class). As a result, the labeler prefers to label normal samples as the positive class, $n_{pl} + n_{pu} \gg n_{nu}$. At the same time, to reduce labeling costs, only a small proportion of the majority class samples are labeled, leading to class-imbalance between labeled and unlabeled samples, $n_l = n_{pl} \ll n_{pu} < n_{pu} + n_{nu} = n_u$. In this case, the methods proposed in Scenario I will no longer be applicable, and a new learning framework needs to be designed.

The problem is depicted in Fig. 3. Compared to Scenario I, the problem in Scenario II is more complex. Labeled normal samples are mixed with the vast unlabeled normal samples, and all of the fault samples are hidden in the unlabeled dataset. Therefore, it is not possible to construct a rough classification boundary by separating labeled samples from the unlabeled dataset as in Scenario I. The main design idea of this framework is to use labeled normal samples to optimize the initial classification boundary, and to adjust the feature mapping and boundary based on the classification results of the unlabeled set by the initial boundary. Hence, the objective function of the model is presented as follows:

$$\min_{\mathbf{W}, R, c} R^2 + \frac{1}{n_l + 1} \sum_{i=1}^{n_l} \max\{0, \|\phi(x_i; \mathbf{W}) - c\|^2 - R^2\} + \hat{R}_U(\mathbf{W}, R, c) + \lambda/2 \sum_{l=1}^L \|\mathbf{W}^l\|^2 \quad (9)$$

$$\hat{R}_U(\mathbf{W}, R, c) = \frac{1}{n_u + 1} \sum_{i=n_l+1}^{n_l+n_u} (\min\{0, \|\phi(x_i; \mathbf{W}) - c\|^2 - R^2\} - \max\{0, \|\phi(x_i; \mathbf{W}) - c\|^2 - R^2\}) \quad (10)$$

The purpose of this objective function is the same as that of (5), which is to use a small hypersphere to encapsulate normal samples and exclude the faulty samples. The second term penalizes labeled samples that fall outside the hypersphere, thereby wrapping labeled samples inside the hypersphere as much as possible to optimize the classification boundary. By the current

boundary of the model, the unlabeled dataset can be divided into untrusty normal samples and untrusty fault samples. The third term $\hat{R}_U(\mathbf{W}, R, c)$ is divided into two parts, where the first part minimizes the distance between untrusty normal samples and the center, and the second part maximizes the distance between untrusty fault samples and the center. After this adjustment, the classification boundary will be clearer.

Unlike Scenario I where all normal samples are in the unlabeled dataset, in Scenario II, it is not possible to estimate the radius R using only the unlabeled dataset. When the class prior π_p is known, the order statistics of the distances from the center to all samples in the training set can be used to determine the radius. We set R to satisfy the following equation:

$$[\pi_p(n_l + n_u)] = \sum_{i=1}^{n_l+n_u} \mathbf{1}[\|\phi(x_i; \mathbf{W}) - c\| \leq R]. \quad (11)$$

If π_p is unknown or cannot be estimated accurately, we cannot reasonably estimate the hypersphere radius R by wrapping a specified number of training samples in the hypersphere. Hence, we need to avoid introducing prior knowledge as much as possible in the objective function. However, since all the faulty samples are in the unlabeled dataset in Scenario II, it is impossible to only introduce a center parameter like in Scenario I. When using information from the unlabeled dataset, an additional classification boundary parameter R is still needed to divide the unlabeled dataset into untrusty normal samples and untrusty faulty samples. Let the hypersphere have a greater margin through optimization. We can set the new objective function as

$$\min_{\mathbf{W}, R, c} \frac{1}{n_l + 1} \sum_{i=1}^{n_l} \|\phi(x_i; \mathbf{W}) - c\|^2 + \hat{R}_U(\mathbf{W}, R, c) + \lambda/2 \sum_{l=1}^L \|\mathbf{W}^l\|^2. \quad (12)$$

The first term of the objective function minimizes the distance from all labeled samples to the center c to make normal samples close to the center, avoiding the introduction of parameter R as in (9). According to the case-control assumption, the labeled dataset is randomly sampled from the true distribution, and all the samples in the labeled dataset are normal samples. In the absence of prior knowledge and without considering outliers, the maximum distance from the center to all labeled samples can be directly used as the boundary parameter R

$$R = \max\{\|\phi(x_i; \mathbf{W}) - c\| \mid x_i \in P\}. \quad (13)$$

D. Optimization and Training Strategy

To make the learned features more stable and improve the convergence speed of the model, the proposed framework uses an unsupervised autoencoder structure to pretrain the model before formal training. Existing research proves that unsupervised pretraining can learn a more balanced feature space than supervised learning [28]. The pretrained encoder will directly serve as the initial feature mapping network for the classification model. To facilitate the loading of pretrained model weights for subsequent training, the pretrained encoder uses the same

network structure as the feature mapping network $\mathbf{W}$ of the deep separation network. The decoder $\mathbf{W}'$ is a mirror structure of the encoder. The pretraining objective function is to minimize the root mean square error after feature reconstruction

$$\min_{\mathbf{W}, \mathbf{W}'} \frac{1}{n_l + n_u} \sum_{i=1}^{n_l + n_u} \|x_i - f_{\mathbf{W}'}(f_{\mathbf{W}}(x_i))\|_2^2. \quad (14)$$

The proposed framework applies a stochastic gradient descent-like (SGD-like) method as the optimizer. Due to the different scales of the network parameters $\mathbf{W}$ and the boundary parameter R , synchronously updating them in each iteration may lead to poor optimization consistency and make the model difficult to converge. Therefore, our framework uses an alternating minimization/block coordinate descent method to alternate between optimizing the network parameters $\mathbf{W}$ and the boundary parameter R . When R is fixed, the network parameters $\mathbf{W}$ are trained for several iterations. Then, after several mini-batch iterations, the updated network parameters $\mathbf{W}$ are used to obtain the latest feature mapping, and the radius R is updated. The training strategy of the framework is shown in Algorithm 1.

IV. EXPERIMENTS AND EVALUATIONS

A. Experimental Settings

As an imbalanced classification problem, the accuracy cannot assess and reflect the model's performance well. In this study, four commonly used metrics in imbalanced classification problems, $F1$ -score, G -mean, matthews correlation coefficient (MCC), and area under the receiver operating characteristic curve (AUCPRC), were chosen [11]. These evaluation metrics are defined based on the binary confusion matrix. The specific calculation formulas are as follows:

$$F1\text{-score} = \frac{2TP}{2TP + FP + FN} \quad (15)$$

$$G\text{-mean} = \sqrt{\frac{TP^2}{(TP + FN)(TP + FP)}} \quad (16)$$

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (17)$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{AUCPRC} = \text{Area Under Precision-Recall Curve}. \quad (18)$$

To validate the proposed framework, we compared it with three categories of baseline methods. It is worth noting that to ensure fair comparisons, all selected baseline methods were based on DNNs. The first category is unsupervised learning methods, including deep OC-SVM and deep SVDD [27]. The second category is PU learning methods that do not consider class imbalance, including non-negative positive unlabeled learning (nnPU) [29] and positive unlabeled learning with labeling bias (PULB) [9]. The third category is PU learning methods that

consider class-imbalance, including CSPU [20] and ImPU [21]. In the same case, the feature network structure was set to be the same for all methods. The nnPU, CSPU, and ImPU methods require knowledge of the class prior π_p to complete the training. Scholars have developed methods, such as [22] and [30], to estimate π_p , but their performance on real industrial datasets is not stable. In the subsequent experiments, to ensure the accuracy of the benchmark methods, the true class prior values of the training set were directly given for these methods that require π_p .

We conduct experiments on two industrial fault detection tasks. The high-speed train wheels fault inspection dataset comes from the project cooperation with China's Xi'an Railway Bureau. The original dataset contains a total of 4700 samples, including 4275 wheel samples that do not require maintenance and 425 wheel samples that require maintenance. Each sample is 800×1 profile data, and all features take values in the real number domain. Due to the small size of the dataset, this task can be considered the class-imbalance caused by absolutely sparse minority samples. The wafer map fault detection dataset is a set of open source datasets from [31]. Each sample is a picture of size 52×52 . Each pixel has three possible values: 0 means blank spot, 1 represents normal die that passed the electrical test, and 2 represents broken die that failed the electrical test. The dataset contains 38 015 samples, including 1 normal type and 8 single defects. A sample may have 2–4 mixed types of defect. The sample size for each defect type ranges from 149 to 2000, details can be found in.¹ We hope that the detection model can detect whether a sample has a single defect type, so eight single defect detection datasets are constructed based on the "1 versus all" principle. Due to the large size of the dataset, this task is a typical relatively imbalanced classification problem.

The experimental samples are randomly divided into training set and test set according to the ratio of 4: 1. All experiments are conducted on an hewlett-packard (HP) server with Intel Xeon Gold 6140 CPU, 2 NVIDIA Quadro P5000 GPU, and 96 GB RAM. To avoid randomness in the results, each experiment is independently conducted 10 times, and all results in the table are reported as (mean—standard deviation). If not specifically stated, the class-IR of the subsequent experimental training set is set to IR = 10 : 1 (normal samples: fault samples), and the proportion of labeled samples in the positive class is $c = 30\%$.

B. Case I: High-Speed Train Wheels Fault Inspection

Wheels are one of the key components that affect the safe and stable operation of high-speed trains. The railway operating and management department usually performs wheel inspections at every 200 000 km operation mileage. On-machine measurement systems installed on portal wheelset lathe can collect wheel profile data to evaluate the health condition of the wheels [32]. If the wheel shape is normal, as shown in Fig. 4(a), no maintenance is required. If the wheels are severely worn and the profile data are not circular, as shown in Fig. 4(b), the wheels need to be machined. Each sample is a 800×1 profile data. The training

¹[Online]. Available: <https://github.com/Junliangwangdhu/WaferMap>

TABLE I
PERFORMANCE OF DIFFERENT METHODS ON HIGH-SPEED TRAIN WHEELS FAULT INSPECTION DATASET

Method	Scenario I: Part of fault samples are labeled				Scenario II: Part of normal samples are labeled			
	AUCPRC	F1-score	G-mean	MCC	AUCPRC	F1-score	G-mean	MCC
deep OC-SVM	0.268–0.073	0.316–0.059	0.343–0.033	0.267–0.072	0.270–0.072	0.317–0.059	0.345–0.033	0.271–0.068
deep SVDD	0.245–0.067	0.306–0.057	0.337–0.027	0.237–0.068	0.244–0.064	0.299–0.049	0.329–0.024	0.232–0.067
nnPU	0.090–0.072	0.183–0.048	0.315–0.039	0.048–0.121	0.305–0.376	0.367–0.324	0.464–0.262	0.278–0.380
PULB	0.848–0.055	0.833–0.071	0.839–0.063	0.819–0.083	0.694–0.117	0.669–0.103	0.674–0.100	0.618–0.115
CSPU	0.767–0.053	0.752–0.043	0.755–0.045	0.734–0.050	0.601–0.089	0.612–0.083	0.614–0.084	0.527–0.095
ImPU	0.889–0.039	0.854–0.027	0.857–0.026	0.844–0.030	0.572–0.093	0.598–0.085	0.603–0.085	0.519–0.096
deepImSN (with π_p)	0.789–0.048	0.760–0.047	0.762–0.047	0.735–0.052	0.915–0.053	0.895–0.027	0.897–0.027	0.884–0.032
deepImSN (without π_p)	0.951–0.010	0.919–0.014	0.920–0.014	0.912–0.015	0.926–0.029	0.888–0.033	0.890–0.032	0.879–0.035

The bold values show the best results among all methods.

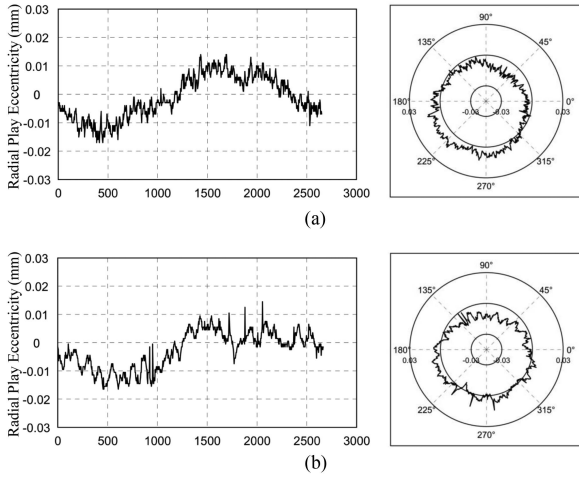


Fig. 4. Measurement of the wheels' profile. The left figures show the radial runout data. The right figures show the real profile of the wheel after visualization, where the radius of the middle circle represents the average radius. (a) Normal sample. (b) Fault sample.

set contains a total of 3420 normal samples, and the number of fault samples and labeled samples are determined based on the IR and labeling ratio (c). In this experiment, we need to apply profile data training a detection model to determine whether the wheels need maintenance.

Table I summarizes the overall performance of different methods on this task. The unsupervised learning methods, deep OC-SVM and deep SVDD, do not perform well due to the lack of guidance from labeling information, and the scores for all evaluation metrics are below 0.35. nnPU does not consider the impact of class-imbalance during the training process, so it has the worst performance among all PU methods. PULB is an indirect PU detection method based on homosequence and is less affected by class imbalance. It has the best overall performance among all baseline methods. When designing the learning objectives, both CSPU and ImPU assume that only some minority class samples are labeled, so their AUCPRC under Scenario I can reach 0.767 and 0.889, respectively. However, when only some majority class samples are labeled in scenario II, the performance of CSPU and ImPU drops significantly, and the evaluation metrics only reach about 0.6. Our proposed deepImSN framework achieved the best performance in both scenarios. Compared with the best-performing baseline method, the performance of the proposed deepImSN is improved by more than 5% in Scenario I and even more than 20% in Scenario II. The p -values of paired t -tests between deepImSN and baseline

TABLE II
 p -VALUES OF THE PAIRED T -TEST BETWEEN BASELINE METHODS AND DEEPIMSN

Method	deepImSN (with π_p)	deepImSN (without π_p)
p -values		
deep OC-SVM	3.49×10^{-7}	2.92×10^{-9}
deep SVDD	4.76×10^{-7}	5.26×10^{-9}
nnPU	7.79×10^{-7}	6.77×10^{-6}
PULB	0.211	0.001
CSPU	0.027	4.39×10^{-5}
ImPU	0.205	0.006

methods are given in Table II. Except when using π_p , the test results with PULB and CSPU, in other cases, the p -value is less than 0.05. The proposed framework has at least one method whose test p -values with all baseline methods are much less than 0.05. These results prove that the improvement of the proposed method is significant.

In Scenario I, due to the combined effects of class-imbalance and partial labeling, the number of labeled samples in the training set was only around 100, making it difficult to accurately estimate the classification boundary R using π_p . Therefore, the performance of deepImSN (with π_p) is less than 0.8. However, deepImSN (without π_p) avoids boundary estimation and achieves the best performance, and the AUCPRC is higher than 0.95. In Scenario II, all baseline methods do not consider the situation where only part of the majority class samples are labeled. Therefore, the scores of these baseline methods are all lower than 0.7. The proposed framework takes this imbalanced scenario into consideration when setting the learning objective, uses PU labels for supervised learning to construct the initial classification boundary, and then makes the boundary clearer through optimization. In this way, unlabeled minority class samples can be identified as much as possible, and the classification performance can reach about 0.9.

To investigate the robustness of the proposed framework under different IRs, we adjusted the number of fault samples in the training set to set the IR to 1, 5, 10, and 20, respectively. The experimental results are shown in Fig. 5. In the case of class balance, all PU learning methods perform well, with AUCPRC close to 100% on the test set. However, as the IR increased, the advantage of the proposed deepImSN method begins to show. Other PU learning methods show a significant decrease in performance as the IR increased, while deepImSN achieved stable performance even at higher IRs. Particularly, the deepImSN method without π_p always achieves the best performance under different experimental conditions, demonstrating the robustness of the proposed framework. When the IR reaches 20, the AUCPRC

TABLE III
OVERALL PERFORMANCE OF DIFFERENT METHODS ON 8 TYPES OF WAFER MAP FAULT DETECTION DATASETS

Method	Scenario I: Part of fault samples are labeled				Scenario II: Part of normal samples are labeled			
	AUCPRC	F1-score	G-mean	MCC	AUCPRC	F1-score	G-mean	MCC
deep OC-SVM	0.191–0.284	0.281–0.257	0.392–0.213	0.091–0.158	0.191–0.284	0.281–0.257	0.392–0.213	0.091–0.158
deep SVDD	0.192–0.268	0.275–0.242	0.386–0.198	0.078–0.134	0.192–0.268	0.275–0.242	0.386–0.198	0.078–0.134
nnPU	0.192–0.234	0.292–0.258	0.398–0.212	0.114–0.322	0.192–0.241	0.289–0.263	0.391–0.214	0.114–0.322
PULB	0.825–0.288	0.857–0.237	0.873–0.197	0.853–0.239	0.533–0.304	0.570–0.247	0.597–0.213	0.506–0.270
CSPU	0.793–0.187	0.815–0.159	0.837–0.146	0.813–0.185	0.517–0.305	0.526–0.279	0.564–0.238	0.477–0.265
ImPU	0.832–0.311	0.869–0.239	0.884–0.198	0.861–0.240	0.528–0.320	0.548–0.293	0.602–0.269	0.495–0.303
deepImSN (with π_p)	0.693–0.181	0.806–0.098	0.814–0.095	0.735–0.217	0.791–0.109	0.806–0.108	0.812–0.103	0.753–0.150
deepImSN (without π_p)	0.932–0.065	0.887–0.077	0.889–0.075	0.879–0.081	0.702–0.356	0.694–0.305	0.716–0.270	0.661–0.336

The bold values show the best results among all methods.

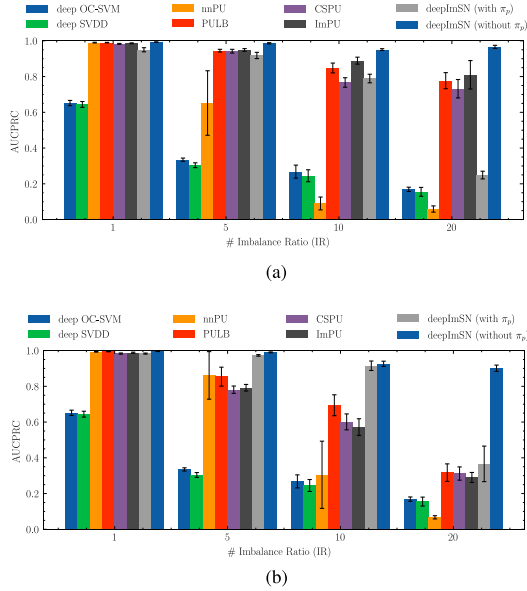


Fig. 5. Performance of methods under difference IR. (a) Scenario I: Part of fault samples are labeled. (b) Scenario II: Part of normal samples are labeled.

of deepImSN is still higher than 0.95 in Scenario I, while the AUCPRC of the baseline methods have dropped below 0.8. deepImSN with π_p also performs well in some IR settings, but this method requires a sufficient number of labeled samples in the training set to accurately estimate the boundary and achieve a good performance. In scenario II, the labeled samples are the majority class, and the AUCPRC of the proposed method always exceeds 0.9 when $IR \leq 10$. Therefore, we recommend using deepImSN without π_p when the number of labeled samples in the training set is limited, and trying deepImSN with π_p when there are more labeled samples available.

The labeled proportion (c) is also an important parameter for PU datasets, which reflects the amount of label information in the training set. When $c = 100\%$, the PU learning problem becomes a supervised learning problem. We conducted experiments by adjusting the number of labeled samples in the training set to set the labeled proportion to 10%, 30%, 50%, and 100%, and the experimental results are summarized in Fig. 6.

When the labeled proportion was 10%, all methods had mediocre performance due to the sparse label information, and deepImSN still achieved the best performance. Compared with the baseline methods, the improvement is about 10%. As the labeled proportion increased, the amount of supervised information in the training set increased, and the performance of baseline methods such as PULB and ImPU improved. The

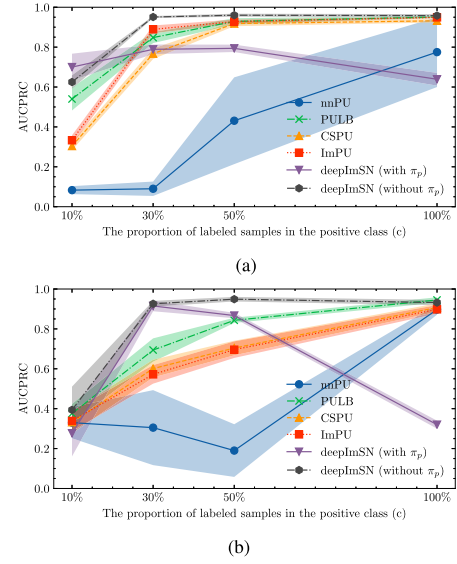


Fig. 6. Performance of PU learning methods under difference labeling ratio. (a) Scenario I: Part of fault samples are labeled. (b) Scenario II: Part of normal samples are labeled.

proposed deepImSN without π_p outperformed baseline methods at any labeled proportion, and it also performed well in the supervised learning setting, demonstrating the superiority of the proposed method. When the labeled proportion of the training set exceeds 50%, the AUCPRC of the PU learning method except nnPU is higher than 0.9. On the other hand, deepImSN with π_p was not suitable for the supervised learning scenario due to the adjustment of the optimization objective, so its performance would decrease when all the training set samples were labeled.

C. Case II: Wafer Map Fault Detection

Defect pattern recognition is essential for determining the root cause of wafer map production defects. The wafer maps were obtained by testing the products in a wafer manufacturing plant [31]. The results are shown in Fig. 7, which includes 8 types of defects. It should be noted that a single wafer may exhibit multiple types of defects simultaneously, as shown in Fig. 7(b). Each sample is test with a 52×52 die. The original training set contains 30 412 samples, which will be used to construct the experimental PU training set based on the IR and labeled proportion. The IR of the training set for different defect types ranges from 3 to 254, and the labeled proportion is always set to 30%. In this experiment, we need to use the wafer maps to train a detection model to determine whether a wafer exhibits a certain type of defect.

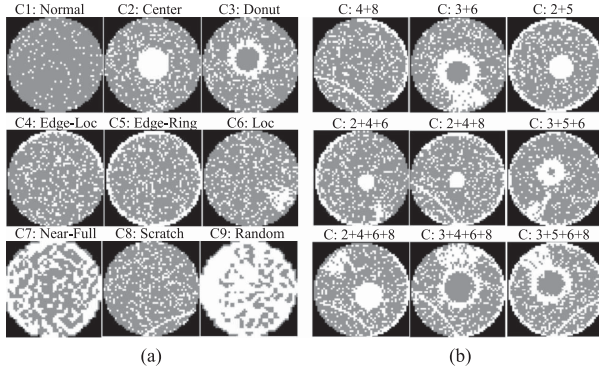


Fig. 7. Visualization of the wafer maps. C1–C9 on the left show nine wafer maps of single-type. A wafer may have multiple types of defects at the same time, and the right shows the examples of nine mixed-types defects. (a) Single-type. (b) Mixed-type.

Due to length constraint, the overall performance of 8 types of defect detection is summarized in Table III (mean-standard deviation). It can be seen that the proposed deepImSN method has the best overall performance among all methods in both scenarios. In Scenario I, compared with well-performing baseline methods such as PULB and ImPU, the deepImSN method has a smaller improvement. Only the AUCPRC has improved by about 10%, and the other three metrics have improved by no more than 3%. This is consistent with the previous analysis of high-speed train dataset. But for scenario II, which is not considered by the baseline methods, the performance improvement of the proposed deepImSN on all metrics exceeds 20%. For Scenario II, the performance of other PU learning methods decreases significantly compared to Scenario I, while the proposed deepImSN method still has satisfactory performance. This fully demonstrates that the proposed framework can be flexibly adjusted for various complex scenarios in practical applications, greatly enhancing the application value of intelligent fault detection methods.

V. CONCLUSION

This article proposes a deepImSN framework for realistic fault detection problem considering class-imbalance and partial label unknown. The framework makes full use of the limited label information in the training set to optimize the feature space and classification boundary more effectively with PU information. By adjusting the optimization objective, the fault detection models can be applied in two scenarios: when only a portion of fault samples are labeled, and when only a portion of normal samples are labeled. The framework adjustment strategy for training set class prior (π_p) known or unknown conditions is also designed. The effectiveness and robustness of the deepImSN framework are verified on two practical problems: high-speed rail wheel fault inspection and wafer map defect detection. The results show the deepImSN is suitable for fault detection tasks in various scenarios and outperforms existing methods.

In the future, we will attempt to further extend the theoretical methods of this framework to multiclassification problems and

apply it to industrial system fault diagnosis tasks to further enhance the practical application value of this method.

APPENDIX

Algorithm 1: The Training Strategy of the DeepImSN Framework.

Input: Training set $T = P \cup U$, initial encoder f_W , initial decoder $f_{W'}$, class prior $\pi_p(\cdot)$, warm-up epoch number m , boundary parameter updating step size k

Output: Trained feature mapping network f_W , center parameter $c(\cdot)$, boundary parameter R

```

1: Set  $Opt$  to be an external SGD-like optimizer
2: Shuffle  $T$  into  $N$  mini-batches, and the  $i$ -th mini-batch is denoted as  $(\mathcal{X}_p^i, \mathcal{X}_u^i)$ 
3: Pretrain autoencoder  $f_W$  and  $f_{W'}$ , use mini-batches to minimize the (14) by  $Opt$ 
4: Calculate the center parameter  $c$  by Eqs.(6);
5: if positive class is the minority class then
6:   if  $\pi_p$  is not None: then
7:     Initialize  $R$  through Eqs.(7);
8:     while no stopping criterion has been met do
9:       for  $i = 1$  to  $N$  do
10:        Set gradient  $\nabla_W \hat{R}_{PU}(\mathcal{X}_p^i, \mathcal{X}_u^i)$  in Eqs.(5)
11:        Update  $W$  by  $Opt$ 
12:        if epoch  $> m$  and  $i\%k == 0$  then
13:          Update  $R$  using Eqs.(7)
14:        end if
15:      end for
16:    end while
17:   else
18:     while no stopping criterion has been met do
19:       for  $i = 1$  to  $N$  do
20:        Set gradient  $\nabla_W \hat{R}_{PU}(\mathcal{X}_p^i, \mathcal{X}_u^i)$  in Eqs.(8)
21:        Update  $W$  by  $Opt$ 
22:      end for
23:    end while
24:   end if
25: end if
26: if positive class is the majority class then
27:   if  $\pi_p$  is not None: then
28:     Initialize  $R$  through Eqs.(11)
29:     while no stopping criterion has been met do
30:       for  $i = 1$  to  $N$  do
31:        Set gradient  $\nabla_W \hat{R}_{PU}(\mathcal{X}_p^i, \mathcal{X}_u^i)$  in Eqs.(9)
32:        Update  $W$  by  $Opt$ 
33:        if epoch  $> m$  and  $i\%k == 0$  then
34:          Update  $R$  using Eqs.(11)
35:        end if
36:      end for
37:    end while
38:   else
39:     Initialize  $R$  through Eqs.(13)
40:     while no stopping criterion has been met do
41:       for  $i = 1$  to  $N$  do
42:        Set gradient  $\nabla_W \hat{R}_{PU}(\mathcal{X}_p^i, \mathcal{X}_u^i)$  in Eqs.(12)
43:        Update  $W$  by  $Opt$ 
44:        if epoch  $> m$  and  $i\%k == 0$  then
45:          Update  $R$  using Eqs.(13)
46:        end if
47:      end for
48:    end while
49:   end if
50: end if

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